YouTube Comments Sentiment Analysis

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GitHub Repository: [Paste your repository URL after uploading]

1. Introduction and Objective

YouTube comments contain opinions and feedback about video content, but reviewing thousands of comments manually is time-consuming. This project aims to analyze up to 2,000 comments from the selected video (https://youtu.be/kAgAQ24_Jso) and classify each as positive, neutral, or negative using Natural Language Processing (NLP). Sentiment charts summarize the distribution.

2. Tools and Methodology

The project uses Python in Visual Studio Code, YouTube Data API v3, Pandas, Matplotlib, and Hugging Face Transformers. The API retrieves published top-level comments in pages of up to 100 items, continuing until 2,000 unique comments are collected or no more are available. Empty comments and duplicate IDs are excluded. The pretrained lxyuan/distilbert-base-multilingual-cased-sentiments-student model assigns each comment a sentiment label and confidence score. Results are exported to CSV and JSON, and bar/pie charts are saved as PNG files.

3. Results (complete after running the program)

Comments analyzed: [actual count]. Positive: [count] ([%]); Neutral: [count] ([%]); Negative: [count] ([%]). Most frequent sentiment: [derive from actual counts]. Insert a generated chart and describe the observed distribution. These fields are placeholders because results must come from a real run, not invented values.

4. Conclusion and Limitations

This project provides a reproducible workflow for collecting comments, estimating sentiment, and visualizing results. Predictions are not ground truth, and confidence scores are not accuracy measurements. The model card lists 12 languages but does not list Urdu, so Urdu/Roman Urdu, code-switching, sarcasm, slang, and emoji-heavy comments may be misclassified. Only top-level comments are collected, and fewer than 2,000 may be available. Future work could manually label a sample and compare models.

References

1. YouTube Data API v3, Comment Threads: List: developers.google.com/youtube/v3/docs/commentThreads/list
2. Hugging Face model card: huggingface.co/lxyuan/distilbert-base-multilingual-cased-sentiments-student